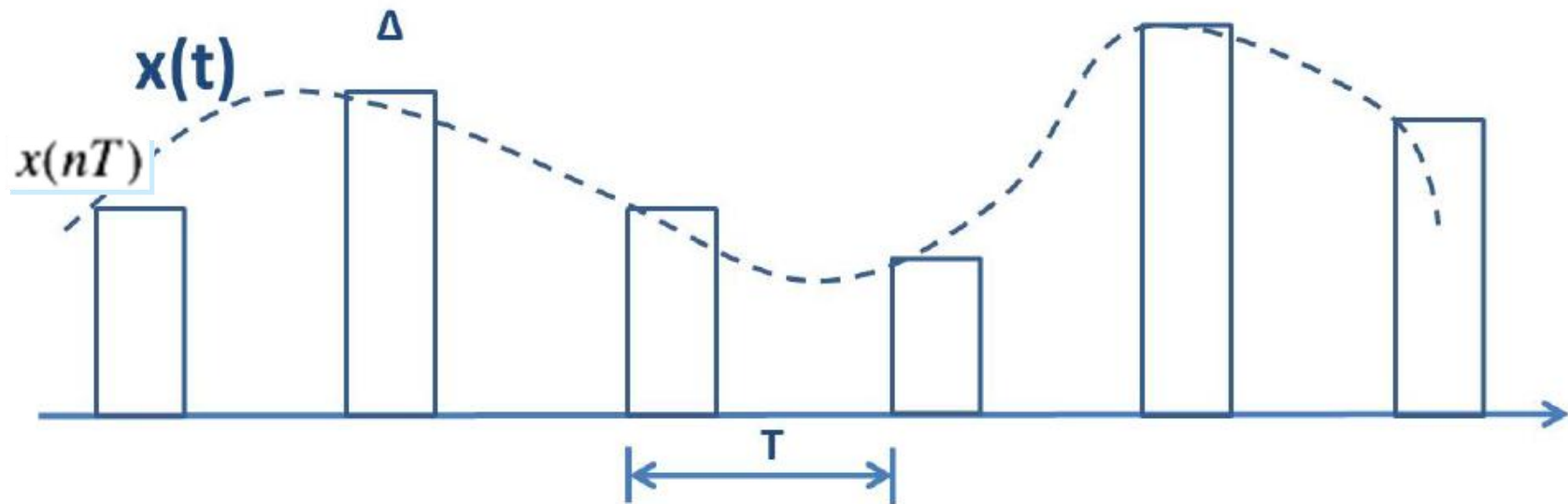


Pulse-Amplitude Modulation (PAM)



- Pulse-Amplitude Modulation (PAM): sample with period T and hold for duration Δ

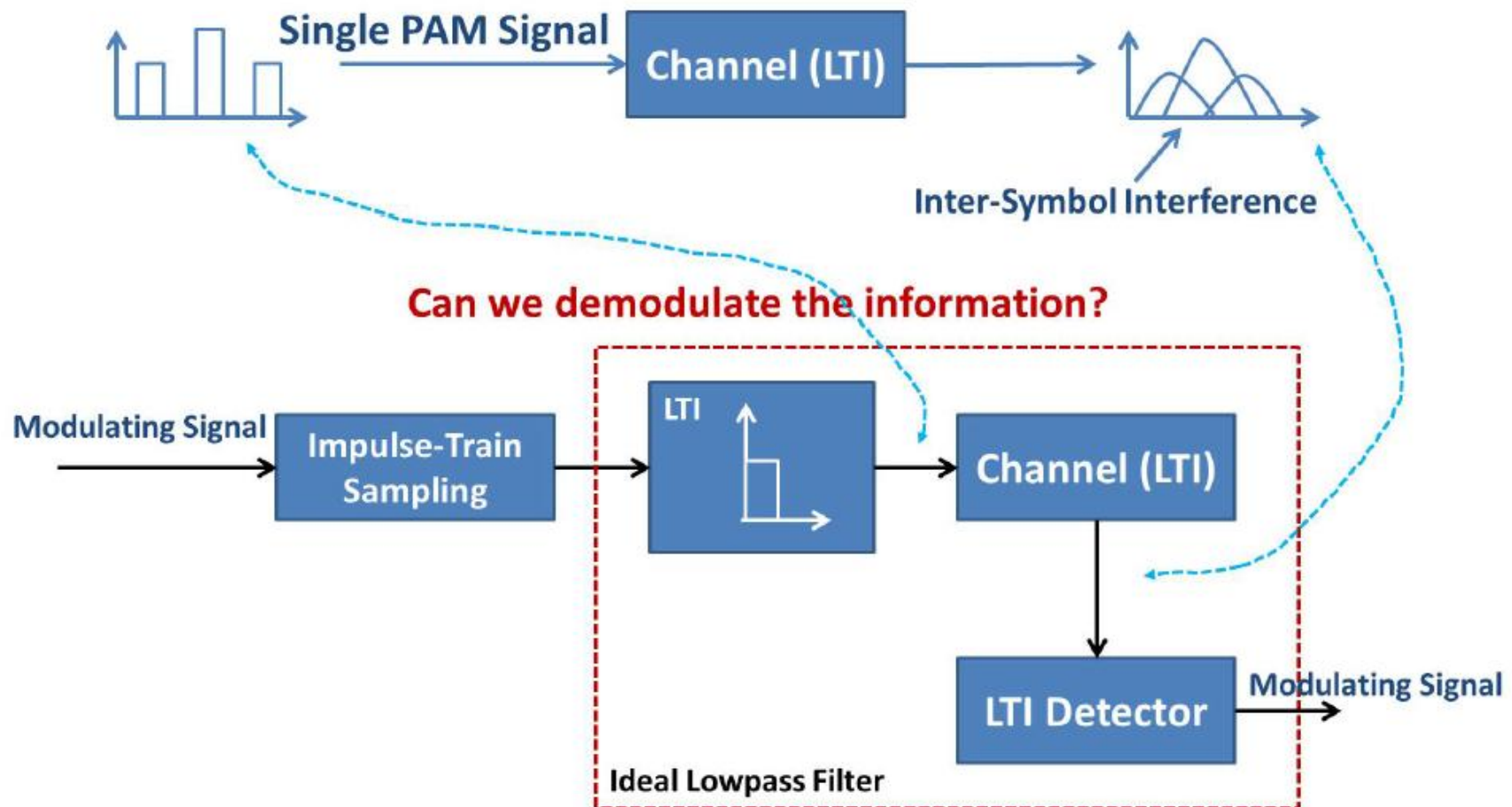
Cont.



Review

Inter-Symbol Interference (ISI)

- Communication channel is usually a low-pass filter
- Receiver recovers the modulating signal with an LTI system

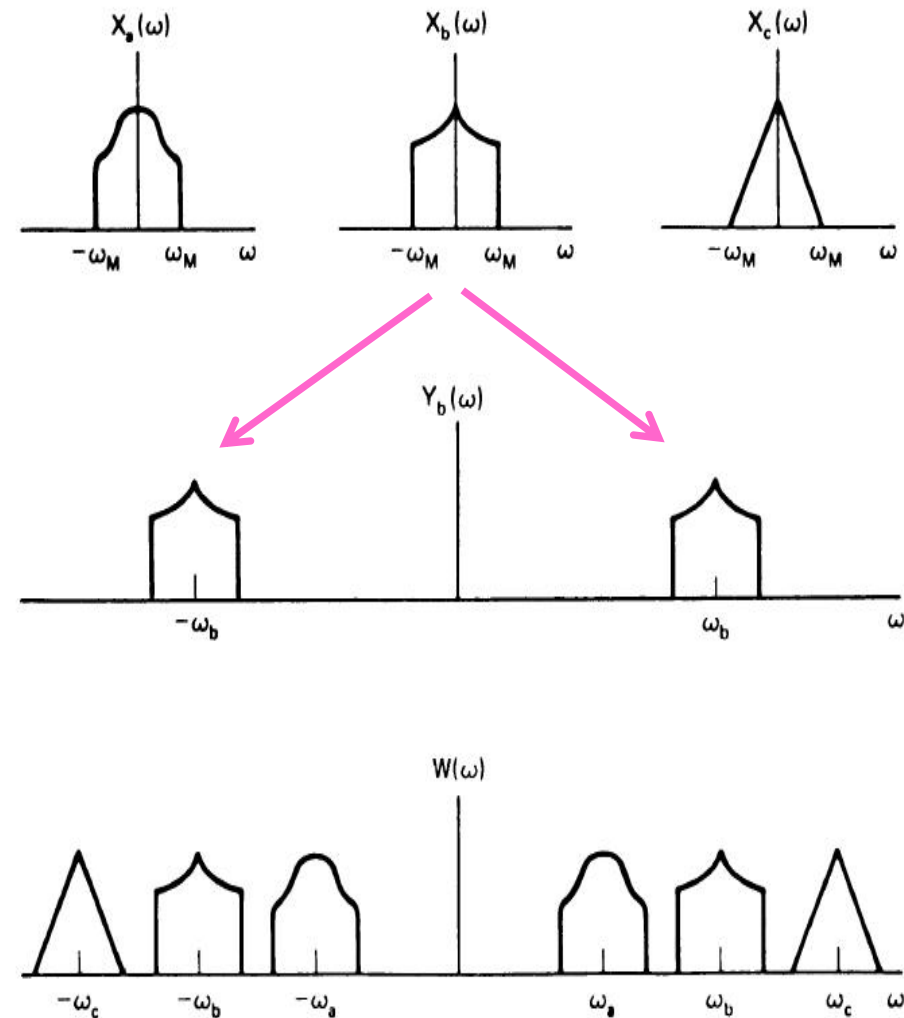
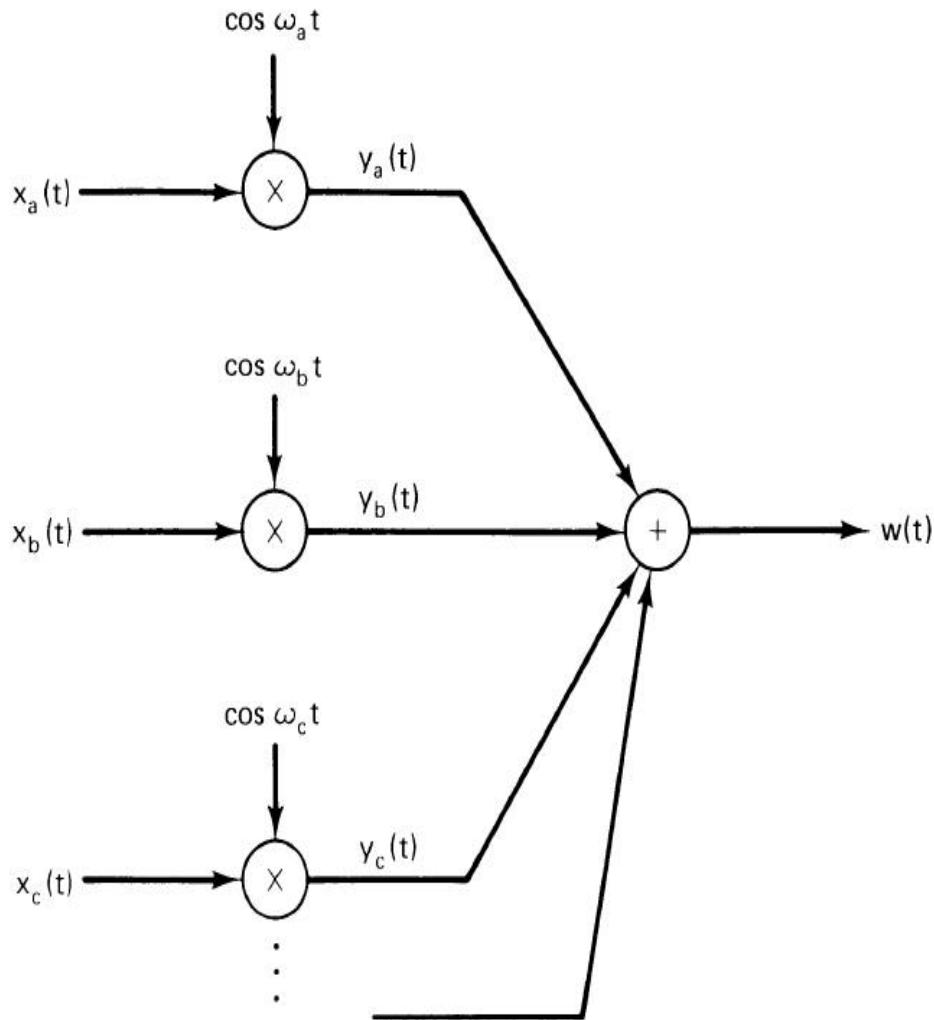


Frequency-division multiplexing (FDM)

- **Systems for transmitting signals provide more bandwidth than is required for one signal.**
 - ◆ E.g., speech signal → 20 ~ 20 kHz
 - microwave channel → 300 MHz ~ 300 GHz
 - satellite link → a few hundred MHz ~ 40 GHz

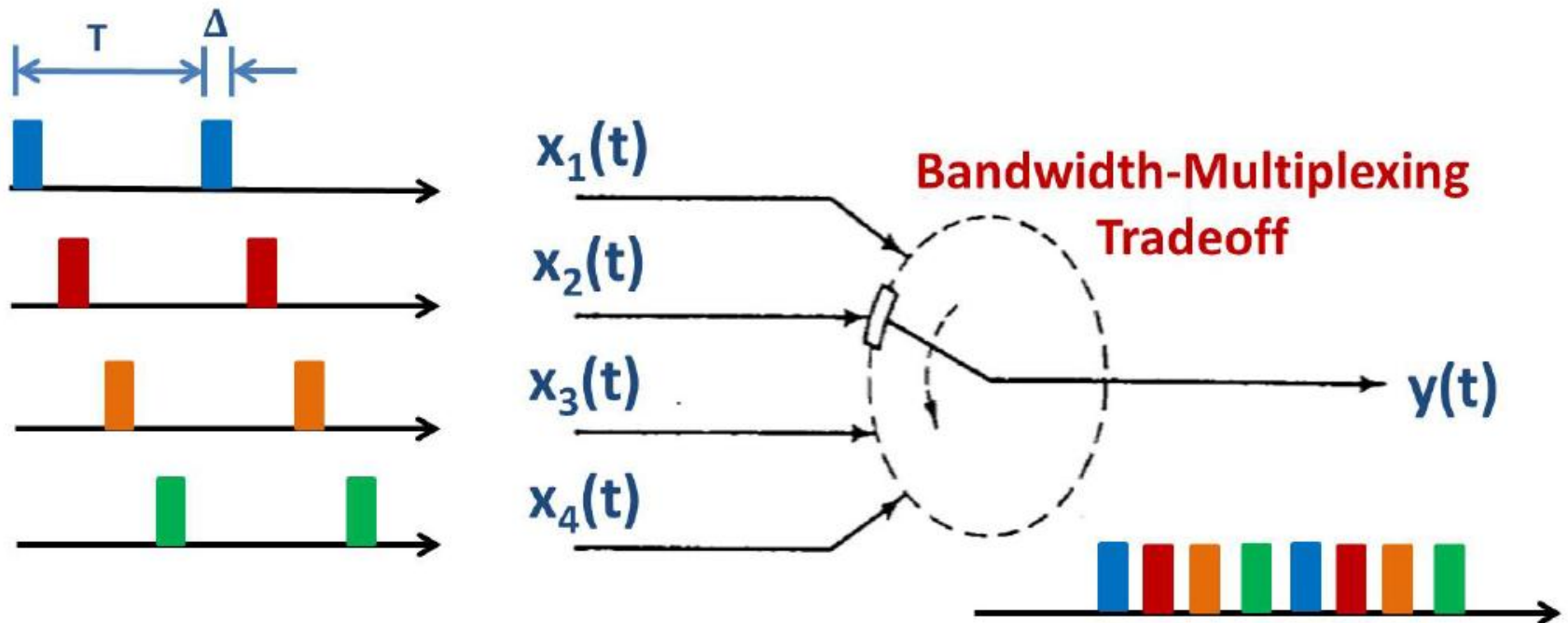
(more in Fig. 8.18)
- **Different modulating signals (e.g., speech), which are overlapping in frequency, can have their spectra **shifted (e.g., by sinusoidal AM) without overlapping**, so they can be transmitted simultaneously over a single wideband channel.**

FDM (cont.)



Time-Division Multiplexing (TDM)

AM signals with pulse-train carrier or PAM signals can be multiplexed in time domain



Angle modulation

$$c(t) = A \cos(\omega_c t + \theta_c) = A \cos \theta(t)$$

$\theta(t) = \omega_c t + \theta_c$ and where ω_c is the frequency and θ_c the phase of the carrier.

Angle modulation:

use modulating signal to change or vary the angle $\theta(t)$

$$y(t) = A \cos[\omega_c t + \theta_c(t)]$$

$$\theta_c(t) = \theta_0 + k_p x(t)$$

Phase modulation: the phase of $y(t)$ is varying with modulating signal

$$y(t) = A \cos \theta(t)$$

$$\frac{d\theta(t)}{dt} = \omega_c + k_f x(t)$$

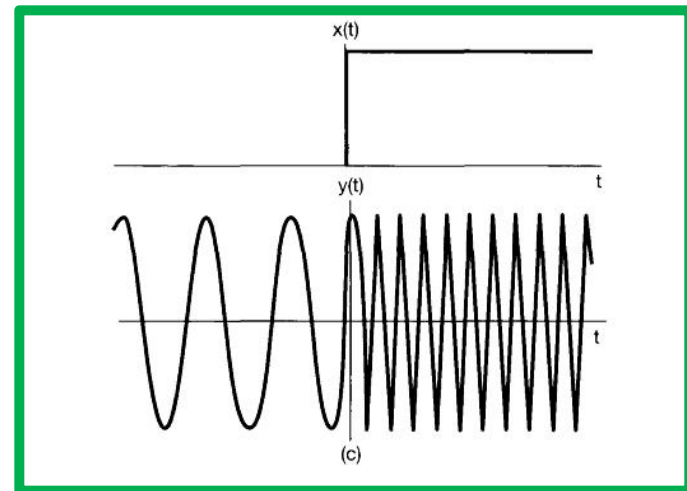
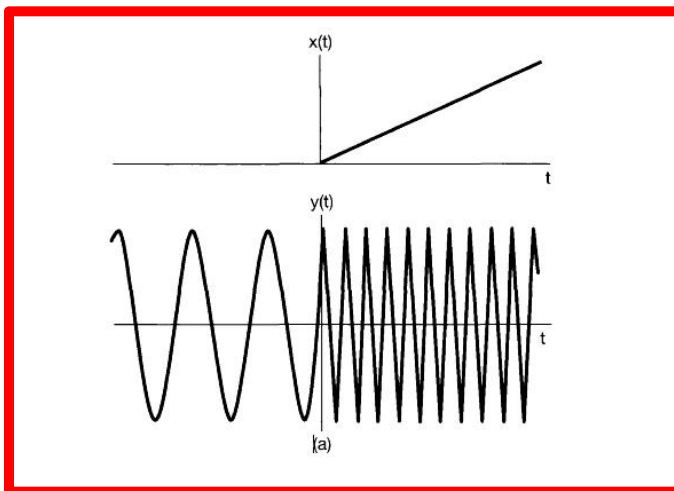
Frequency modulation: the derivative of the angle is varying with modulating signal

$$\frac{d\theta(t)}{dt} = \omega_c + k_p \frac{dx(t)}{dt}$$

$$\frac{d\theta(t)}{dt} = \omega_c + k_f x(t)$$

- **Phase modulating with $x(t)$ is identical to frequency modulating with the derivative of $x(t)$.**
- Likewise, **frequency modulating with $x(t)$ is identical to phase modulating with the integral of $x(t)$.**

Fig. 8.32



Narrowband frequency modulation

$$x(t) = A \cos \omega_m t$$

$$\omega_i(t) = \omega_c + k_f A \cos \omega_m t,$$

which varies sinusoidally between $\omega_c + k_f A$ and $\omega_c - k_f A$.

With $\Delta\omega = k_f A$, we have $\omega_i(t) = \omega_c + \Delta\omega \cos \omega_m t$,

$$y(t) = \cos[\omega_c t + \int x(t) dt] = \cos\left(\omega_c t + \frac{\Delta\omega}{\omega_m} \sin \omega_m t + \theta_0\right),$$

where θ_0 is a constant of integration. For convenience we will choose $\theta_0 = 0$, so that

$$y(t) = \cos\left[\omega_c t + \frac{\Delta\omega}{\omega_m} \sin \omega_m t\right].$$

Modulation index for frequency modulation $m = \Delta\omega/\omega_m$

Cont.

- The properties of FM systems tend to be different, depending on whether the modulation index m is small or large.

$$y(t) = \cos(\omega_c t + m \sin \omega_m t)$$

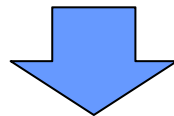
or

$$y(t) = \cos \omega_c t \cos(m \sin \omega_m t) - \sin \omega_c t \sin(m \sin \omega_m t).$$

When m is sufficiently small ($\ll \pi/2$), we can make the approximations

$$\cos(m \sin \omega_m t) \approx 1,$$

$$\sin(m \sin \omega_m t) \approx m \sin \omega_m t,$$



$$y(t) \approx \cos \omega_c t - \frac{m(\sin \omega_m t)(\sin \omega_c t)}{1}$$

↑
carrier

↑
signal

↑
carrier

Cont.

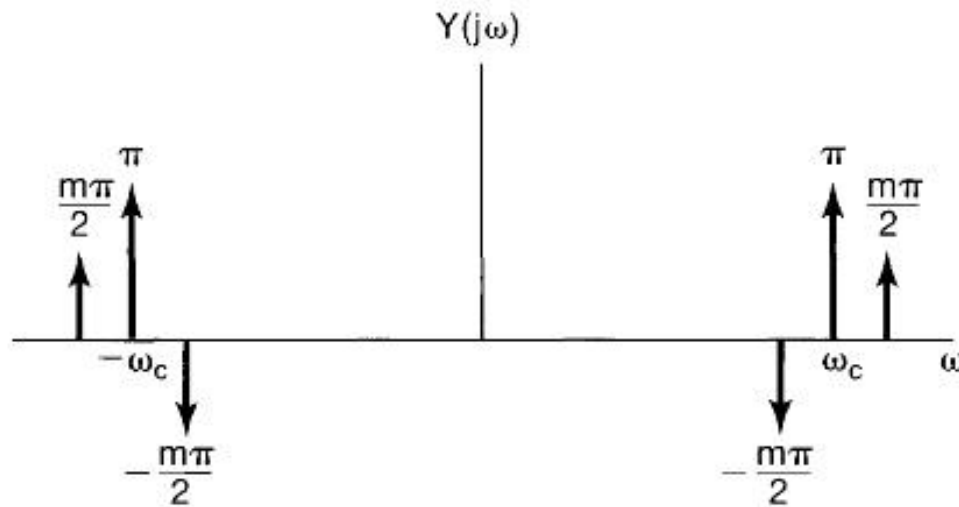


Figure 8.33 Approximate spectrum for narrowband FM.

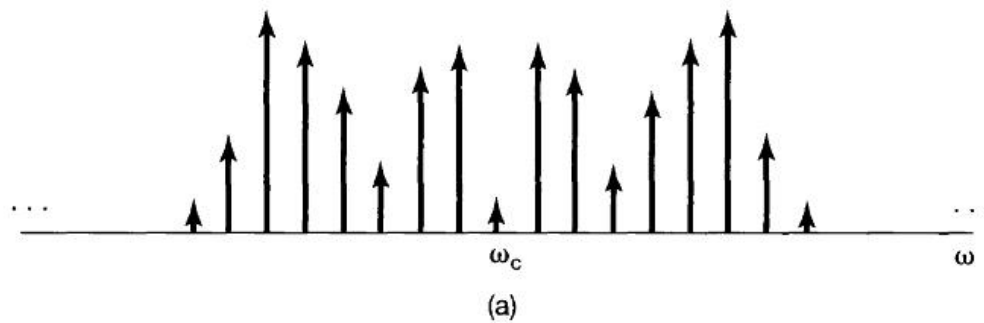
- The bandwidth of the sidebands is equal to the bandwidth of the modulating signal.
 - ✓ The bandwidth of the sidebands is independent of the modulation index m (i.e., it depends only on the bandwidth of the modulating signal, not on its amplitude).

Wideband Frequency Modulation

$$y(t) = \cos \omega_c t \cos(m \sin \omega_m t) - \sin \omega_c t \sin(m \sin \omega_m t).$$

- The terms $\cos[m \sin \omega_m t]$ and $\sin[m \sin \omega_m t]$ are ω_m periodic signals with fundamental frequency
- Thus, the Fourier transform of each of these signals is an impulse train with impulses at integer multiples of ω_m and amplitudes proportional to the Fourier series coefficients.
- The coefficients for these two periodic signals involve a class of functions referred to as **Bessel functions** of the first kind.
 - ◆ The first term corresponds to a sinusoidal carrier of the form $\cos \omega_c t$ amplitude modulated by the periodic signal $\cos[m \sin \omega_m t]$
 - ◆ The second term to a sinusoidal carrier $\sin \omega_c t$ amplitude modulated by the periodic signal $\sin[m \sin \omega_m t]$

Cont.



$$B \simeq 2k_f A = 2\Delta\omega.$$

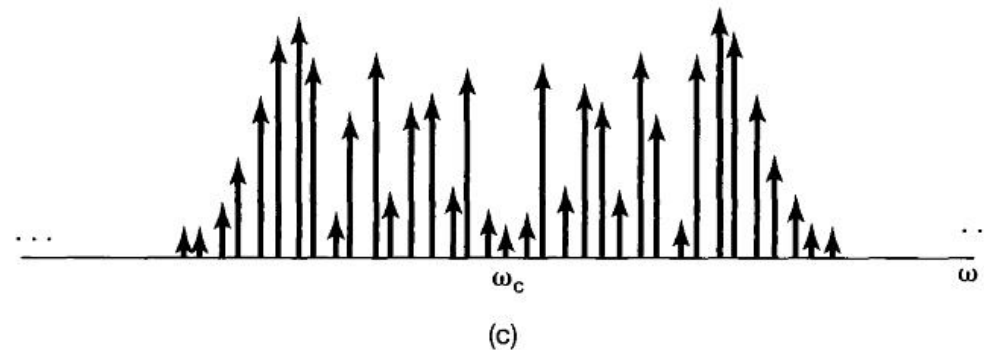
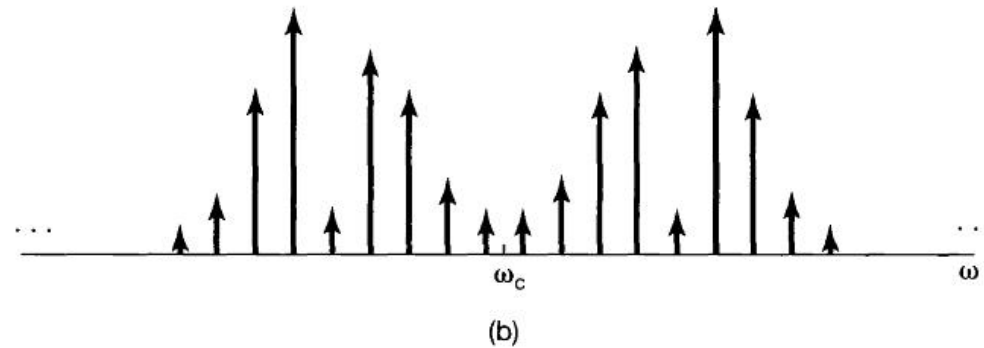


Figure 8.35 Magnitude of spectrum of wideband frequency modulation with $m = 12$: (a) magnitude of spectrum of $\cos \omega_c t \cos[m \sin \omega_m t]$; (b) magnitude of spectrum of $\sin \omega_c t \sin[m \sin \omega_m t]$; (c) combined spectral magnitude of $\cos[\omega_c t + m \sin \omega_m t]$.

Cont.

- For wideband FM, since we assume that m is large, the bandwidth of the modulated signal is much larger than the bandwidth of the modulating signal.
- In contrast to the narrowband case, the bandwidth of the transmitted signal in wideband FM is directly proportional to **amplitude A of the modulating signal and the gain factor k_f .**

Chapter 6

Time & Frequency Characterization of Signals and Systems

Outline

1. Magnitude and phase of Fourier transform
2. Group delay
3. Non-ideal low-pass filter

1. Magnitude & Phase

$$y(t) = A \cos(\omega t + \theta)$$

$$y(t) = \sum_i A_i \cos(\omega_i t + \theta_i)$$

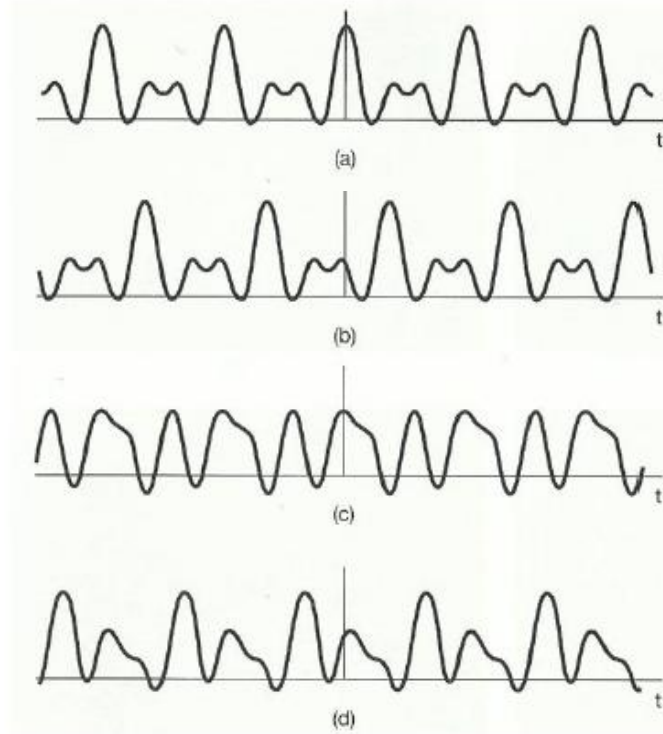


Magnitude & Phase (Cont.)

- Magnitude of spectrum, $|X(j\omega)|$ or $|X(e^{j\omega})|$, determines the energy of frequency component
- Phase of spectrum affects the shape of time-domain signal

$$x(t) = 1 + \frac{1}{2} \cos(2\pi t + \phi_1) + \cos(4\pi t + \phi_2) + \frac{2}{3} \cos(6\pi t + \phi_3).$$

Figure 6.1



$$\phi_1 = \phi_2 = \phi_3 = 0;$$

$$\phi_1 = 4 \text{ rad}, \phi_2 = 8 \text{ rad}, \phi_3 = 12 \text{ rad};$$

$$\phi_1 = 6 \text{ rad}, \phi_2 = -2.7 \text{ rad}, \phi_3 = 0.93 \text{ rad};$$

$$\phi_1 = 1.2 \text{ rad}, \phi_2 = 4.1 \text{ rad}, \phi_3 = -7.02 \text{ rad}.$$

Example 1: Magnitude & Phase of Image

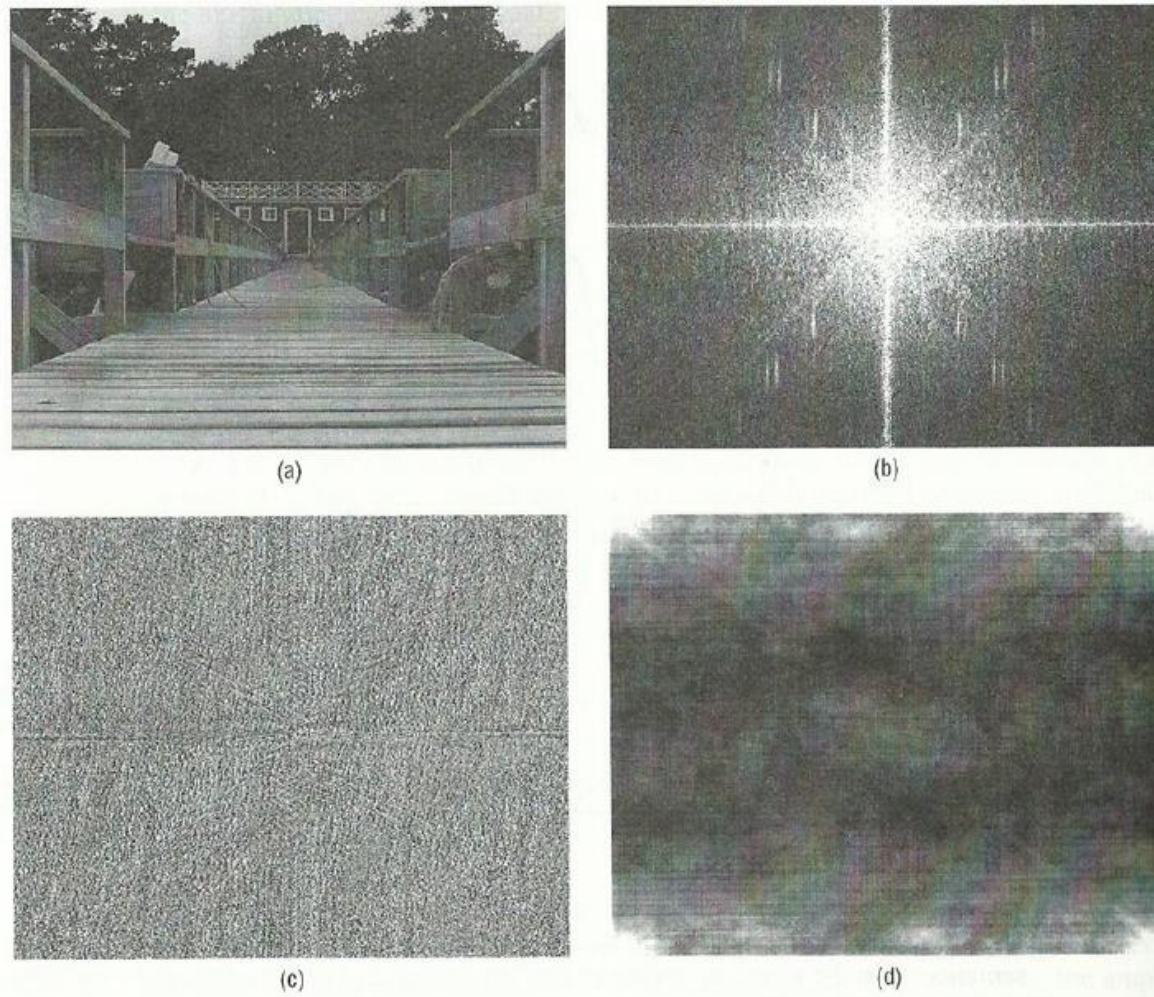
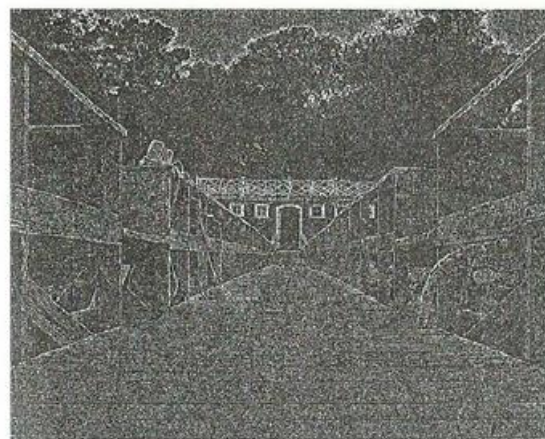


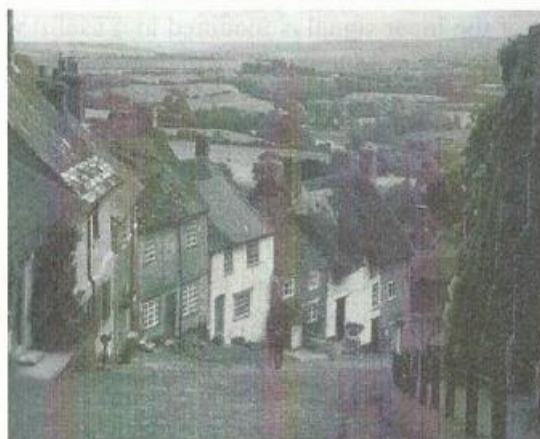
Figure 6.2 (a) Original image; (b) Magnitude; (c) Phase; (d) Set phase to zero



(e)



(f)

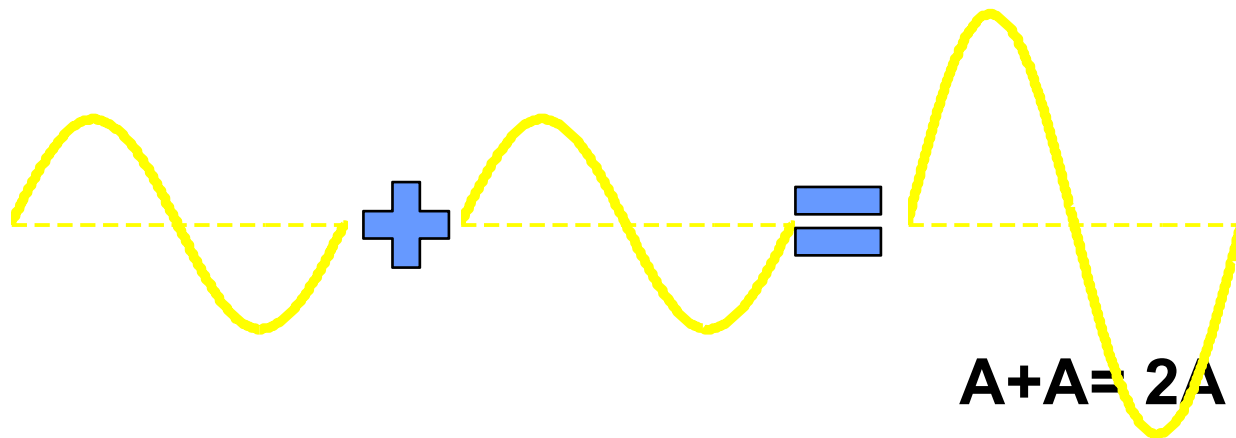


(g)

Figure 6.2 (e) Set magnitude to 1; (f) Original phase + (g)'s magnitude

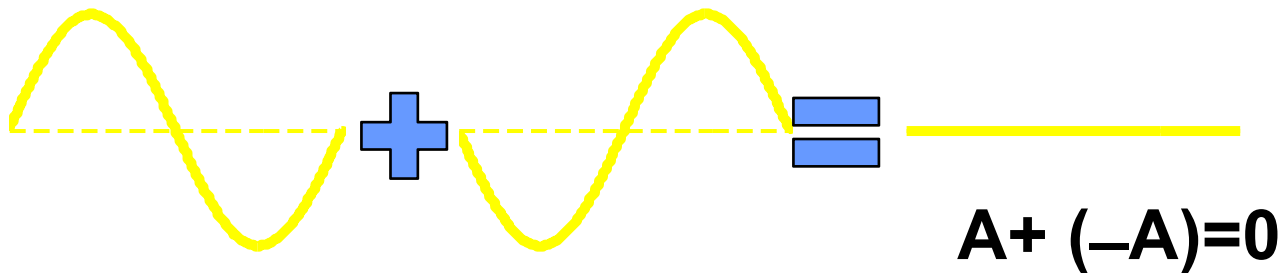
Adding two pure-tones

- When two pure-tones have the same frequency, we have two special cases:
 - 1) Two pure-tones in phase, or phase difference $= 0^\circ$



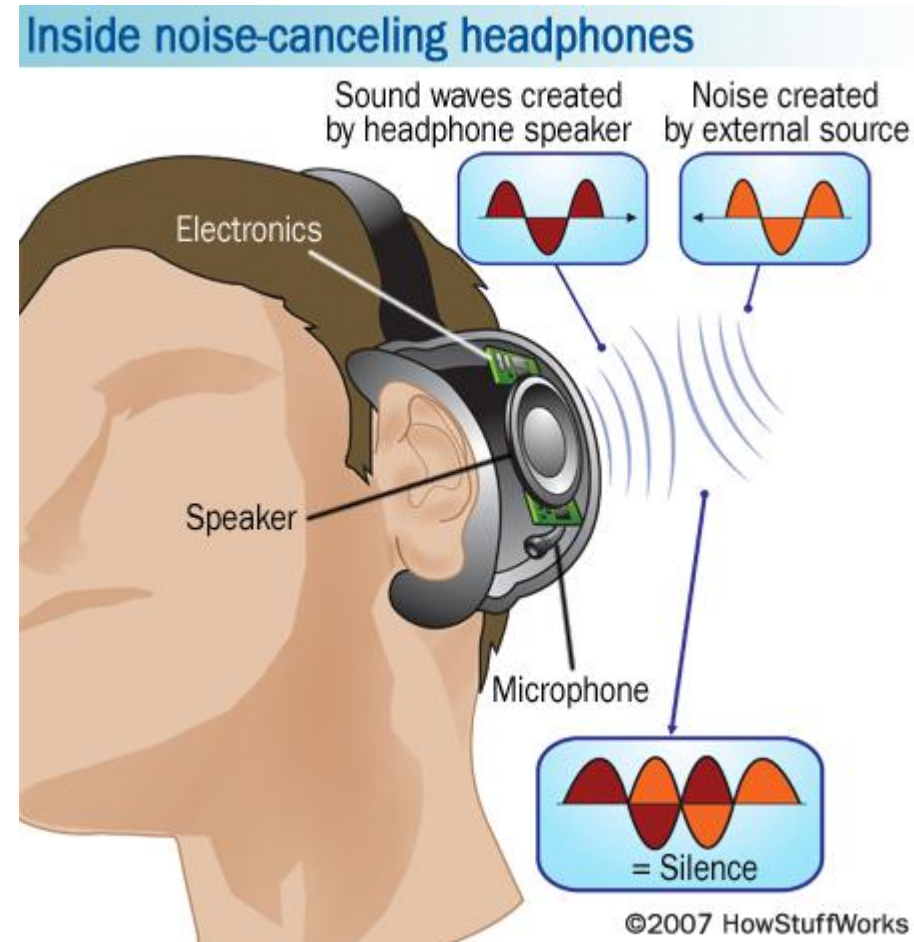
Adding two pure-tones (cont.)

- 2) Two pure-tones out of phase, i.e., phase difference $\neq 0^\circ$
- ◆ Phase difference = $180^\circ \rightarrow$ complete cancellation



Noise-cancelling headphones

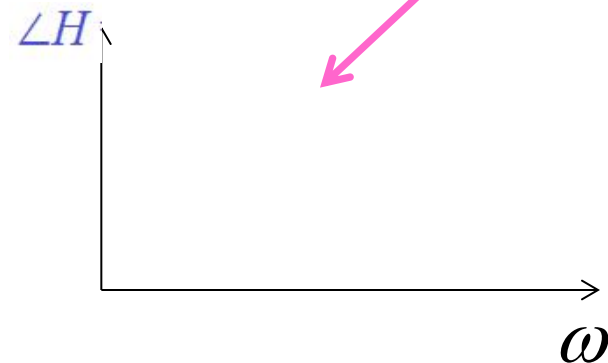
- **Cancel out the noise from background by producing sound waves that cancel the environmental noise.**
 - 1) **microphone to pickup incoming noise,**
 - 2) **processor to determine the frequency of incoming sound, and**
 - 3) **speaker to emit the signal which is exactly out of phase with the noise you want to cancel.**



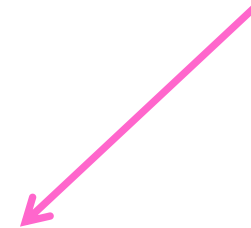
2. LTI Systems: Linear & Nonlinear Phase

$$X(j\omega) \longrightarrow \boxed{H(j\omega)} \longrightarrow Y(j\omega) = H(j\omega)X(j\omega)$$

$$X(e^{j\omega}) \longrightarrow \boxed{H(e^{j\omega})} \longrightarrow Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$



Linear?

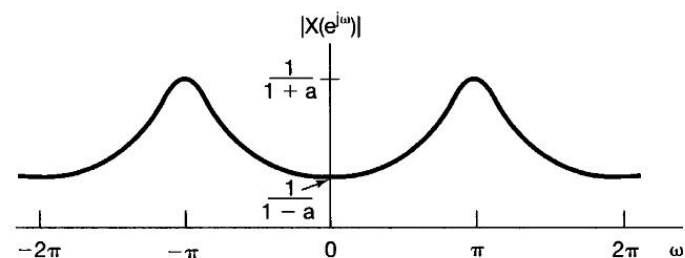
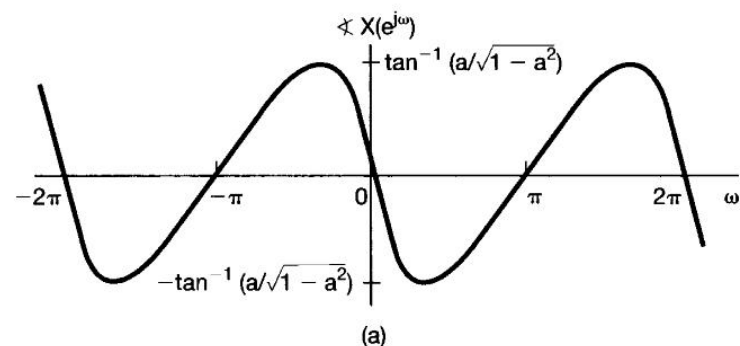
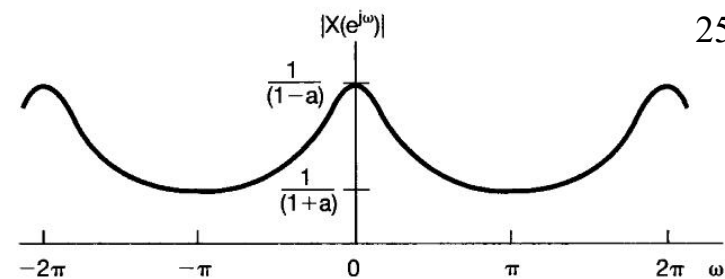


Example 5.1

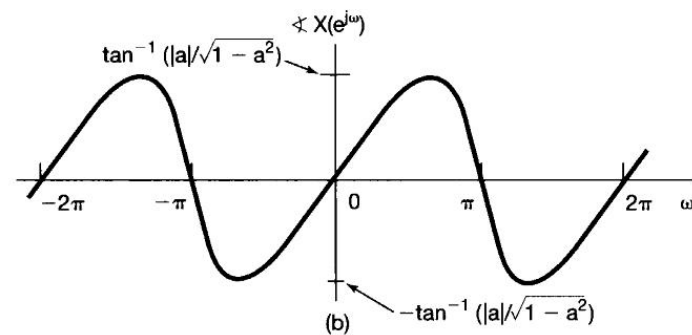
$$x[n] = a^n u[n], \quad |a| < 1.$$

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{+\infty} a^n u[n] e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}}. \end{aligned}$$

$a > 0$



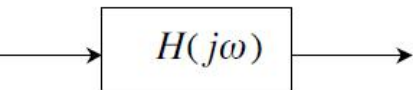
$a < 0$



Group Delay

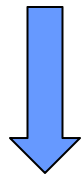
- Linear phase response leads to delay

$$X(j\omega)H(j\omega) = X(j\omega)e^{-j\omega t_0} \longleftrightarrow x(t - t_0)$$



$$H(j\omega) = e^{-j\omega t_0}$$

$$|H(j\omega)| = 1, \quad \angle H(j\omega) = -\omega t_0$$



$$y(t) = x(t - t_0)$$

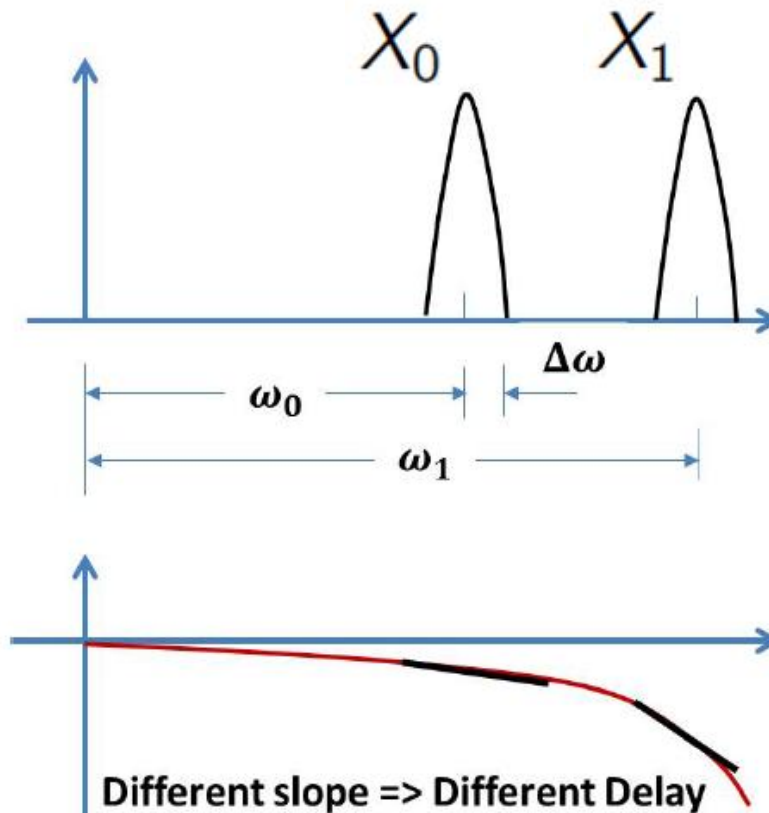
Cont.

- Systems with linear phase characteristics have the particularly simple interpretation as time shifts.
- The phase **slope** tells us the size of the time shift.
 - ◆ If $\angle H(j\omega) = -\omega t_0$, the system imparts a time shift of $-t_0$

Cont.

Navigation icons: back, forward, search, etc.

- Non-linear phase response leads to distortion: $e^{-j\omega^2 t_0}$
- Narrow-band signal: phase response can be approximated as linear



Cont.

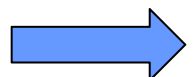
Suppose two narrow-band signals X_0 and X_1 are delivered into a system $H(j\omega) = e^{-j\omega^2 t_0}$:

$$-j\omega^2 t_0 \approx -2j\omega_0 t_0 \omega + j\omega_0^2 t_0$$

$$\begin{aligned} X_0(j\omega)H(j\omega) &= X_0(j\omega)e^{-j\omega^2 t_0} \\ &\approx X_0(j\omega)e^{-2j\omega_0 t_0 \omega} e^{j\omega_0^2 t_0} \\ &\leftrightarrow e^{j\omega_0^2 t_0} x_0(t - 2\omega_0 t_0) \end{aligned}$$

$$X_1(j\omega)H(j\omega) \leftrightarrow e^{j\omega_1^2 t_0} x_1(t - 2\omega_1 t_0)$$

Group delay: different frequency components have different delay

 **waveform distortion**

Example

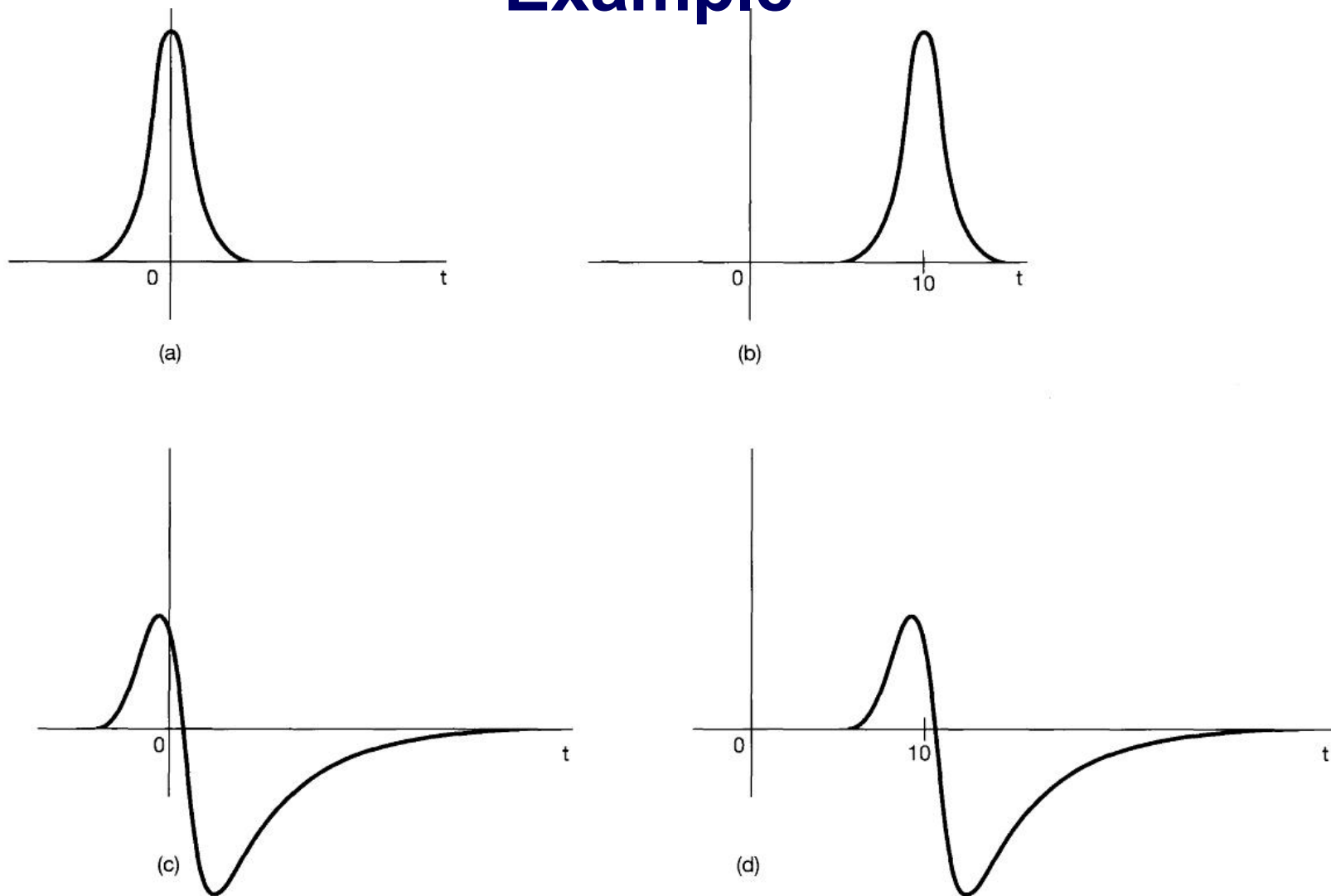


Figure 6.3 (a) Continuous-time signal that is applied as the input to several systems for which the frequency response has unity magnitude; (b) response for a system with linear phase; (c) response for a system with nonlinear phase; and (d) response for a system with phase equal to the nonlinear phase of the system in part (c) plus a linear phase term.

Bode plot

Plots of $20 \log_{10} |H(j\omega)|$ and $\angle H(j\omega)$ versus $\log_{10}(\omega)$

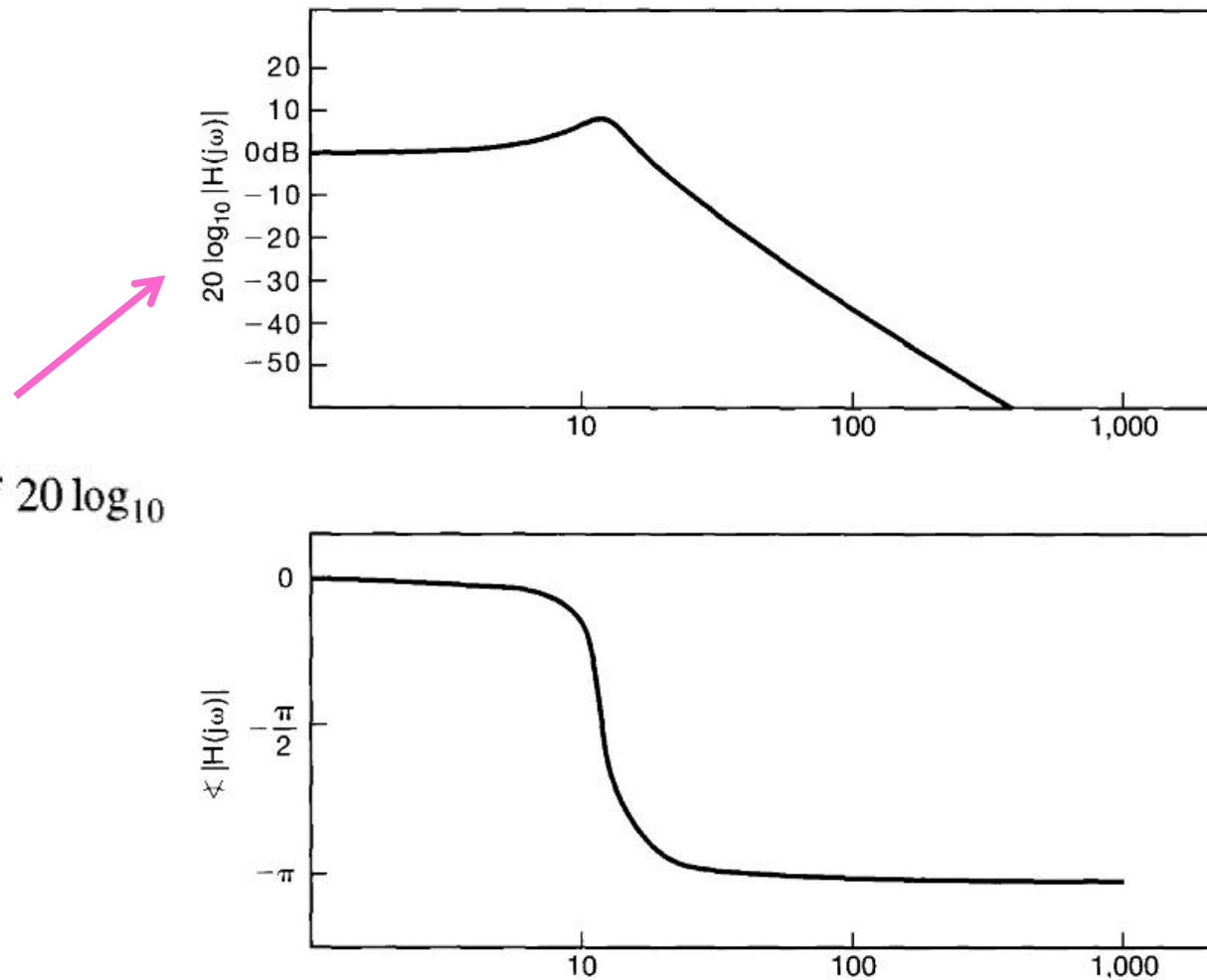
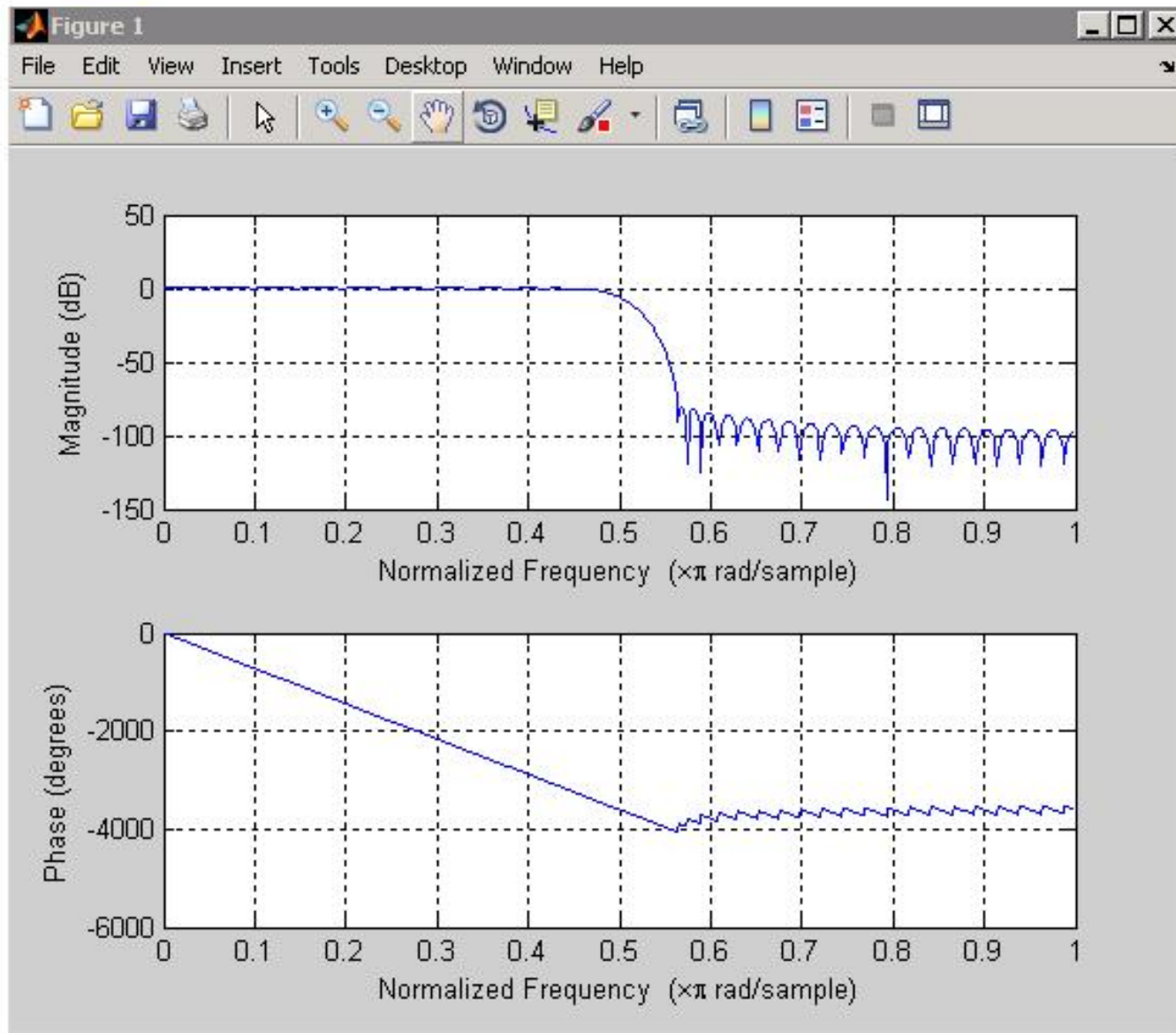


Figure 6.8 A typical Bode plot. (Note that ω is plotted using a logarithmic scale.)

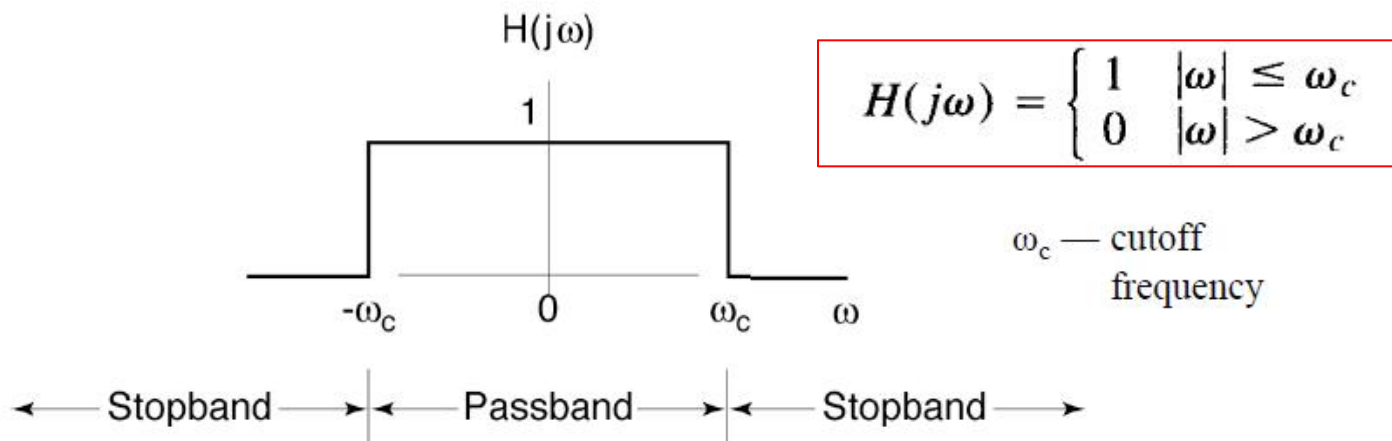
```
b = fir1(80, 0.5, kaiser(81, 8));  
freqz(b, 1);
```



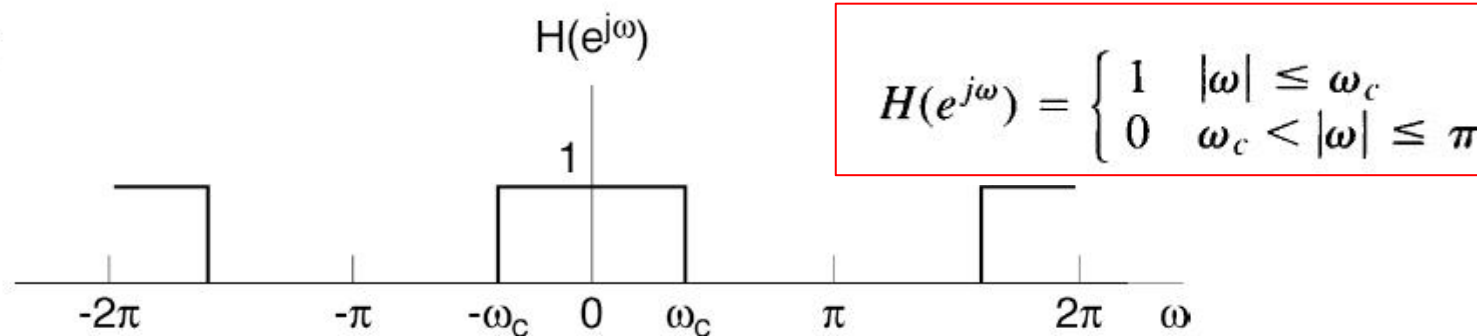
3. Ideal Low-pass Filter

- Ideal low-pass filter: non-causal, sharp edge, implementation issue

CT

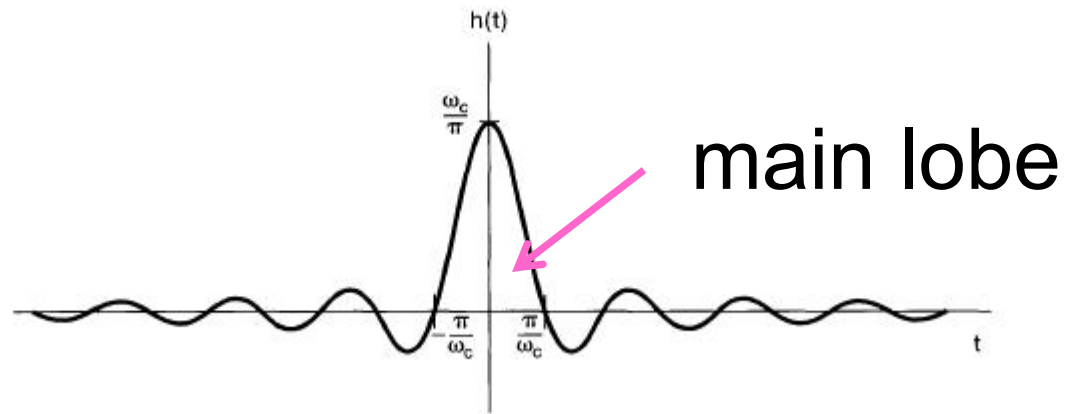


DT

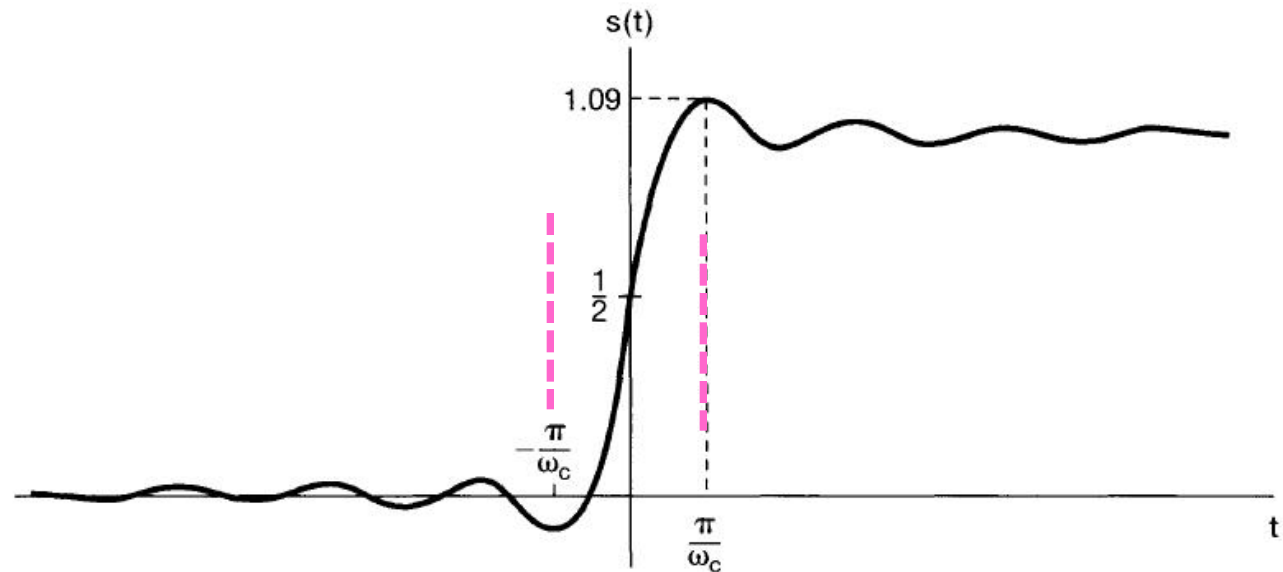
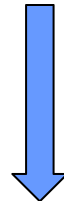


Note: $|H| = 1$ and $\angle H = 0$ for the ideal filters in the passbands,
no need for the phase plot.

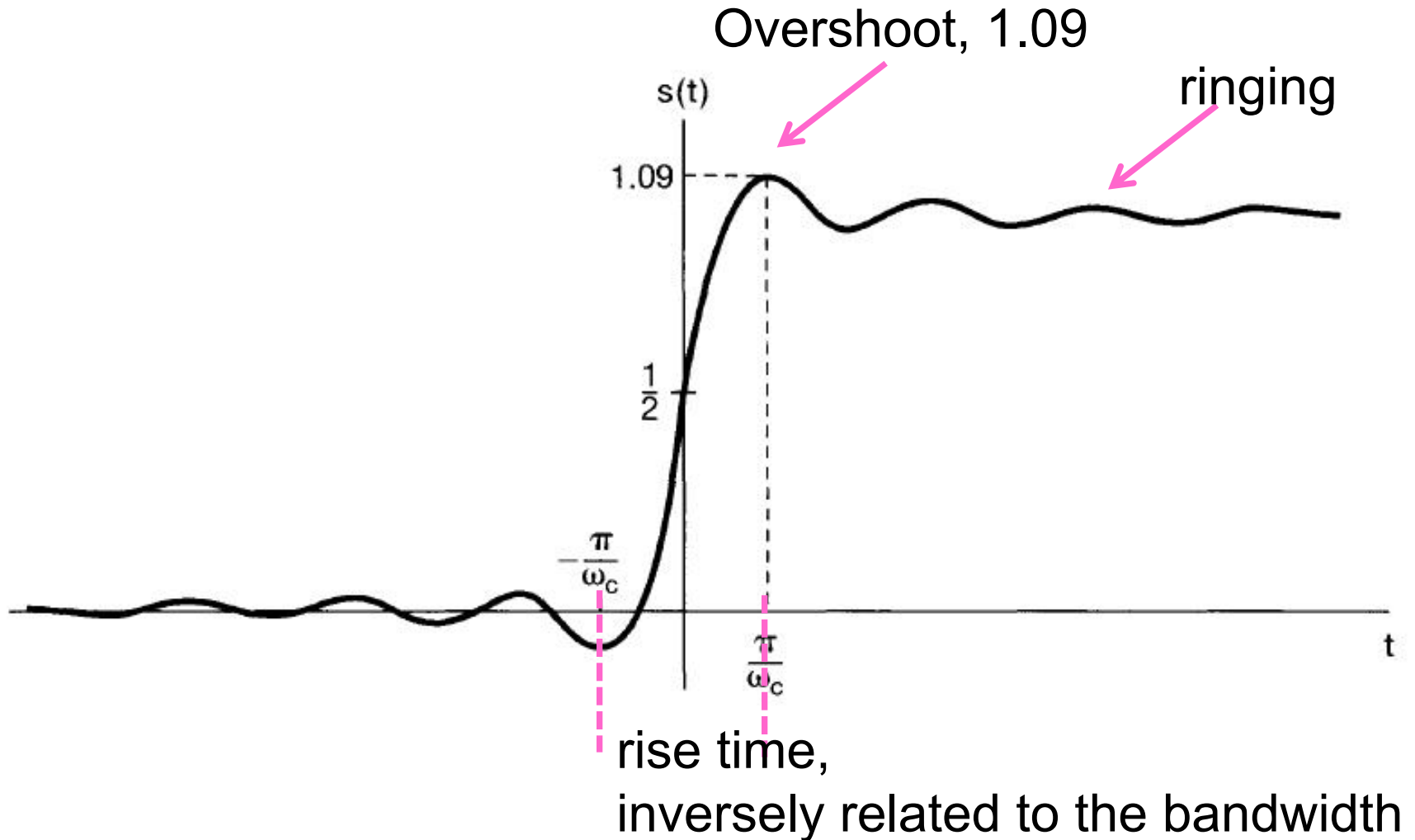
Step response of low-pass filter



$$s(t) = \int_{-\infty}^t h(\tau) d\tau,$$



Cont.



Non-ideal Low-pass Filter

- Non-ideal filter: passband (with tolerant ripple) $\rightarrow$ transition band $\rightarrow$ stopband

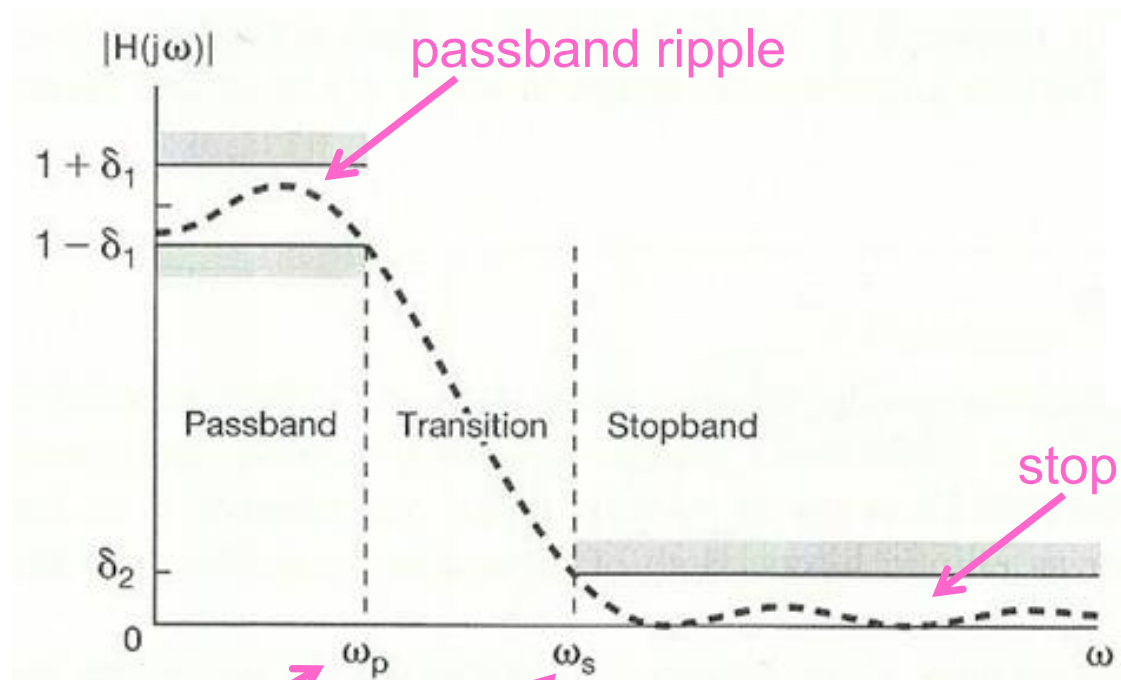


Figure 6.16 Tolerances for the magnitude characteristic of a lowpass filter. The allowable passband ripple is δ_1 and stopband ripple is δ_2 . The dashed curve illustrates one possible frequency response that stays within the tolerable limits.

passband edge stopband edge

Cont.

- Step response: rise time, overshoot, ringing frequency, settling time

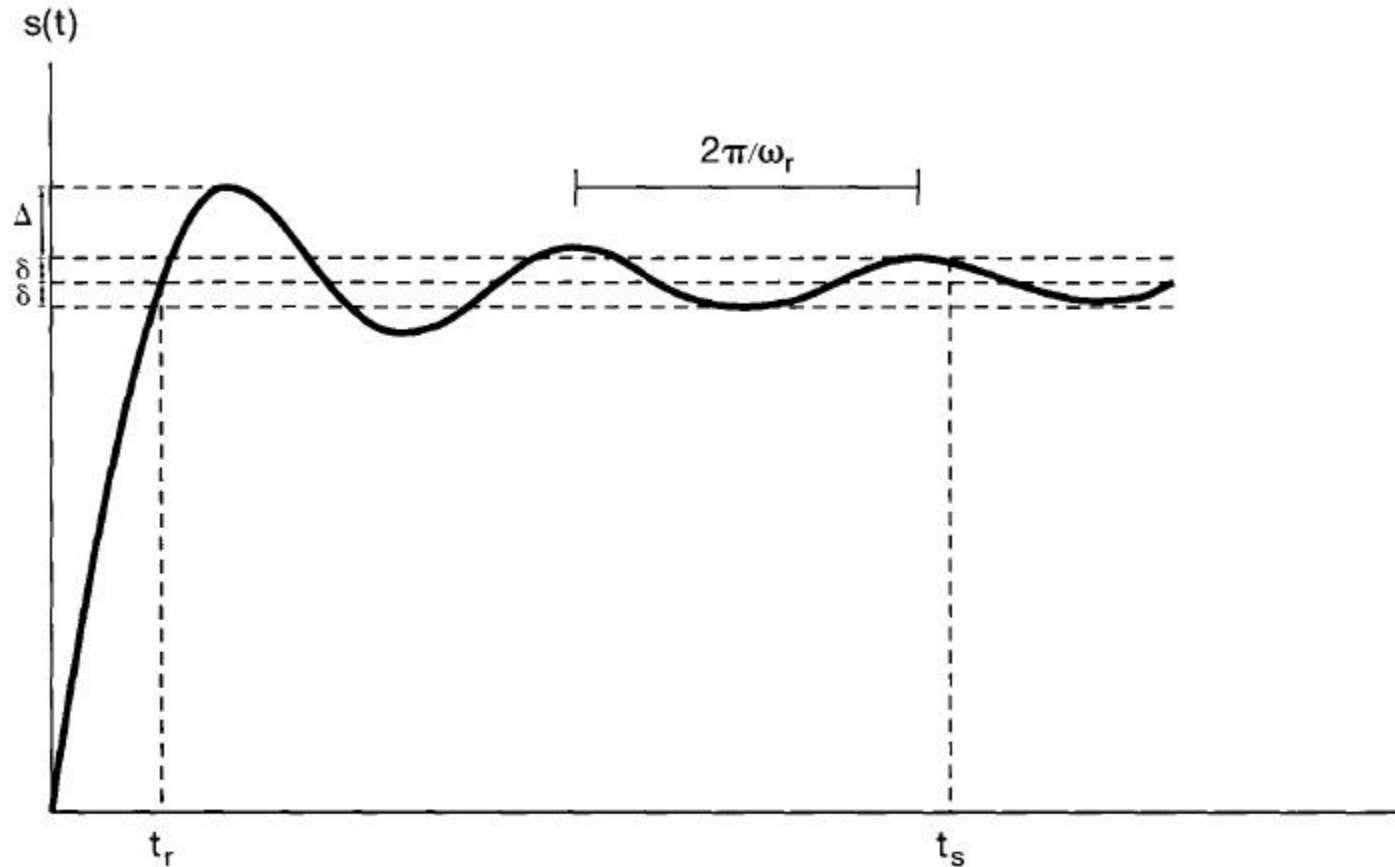
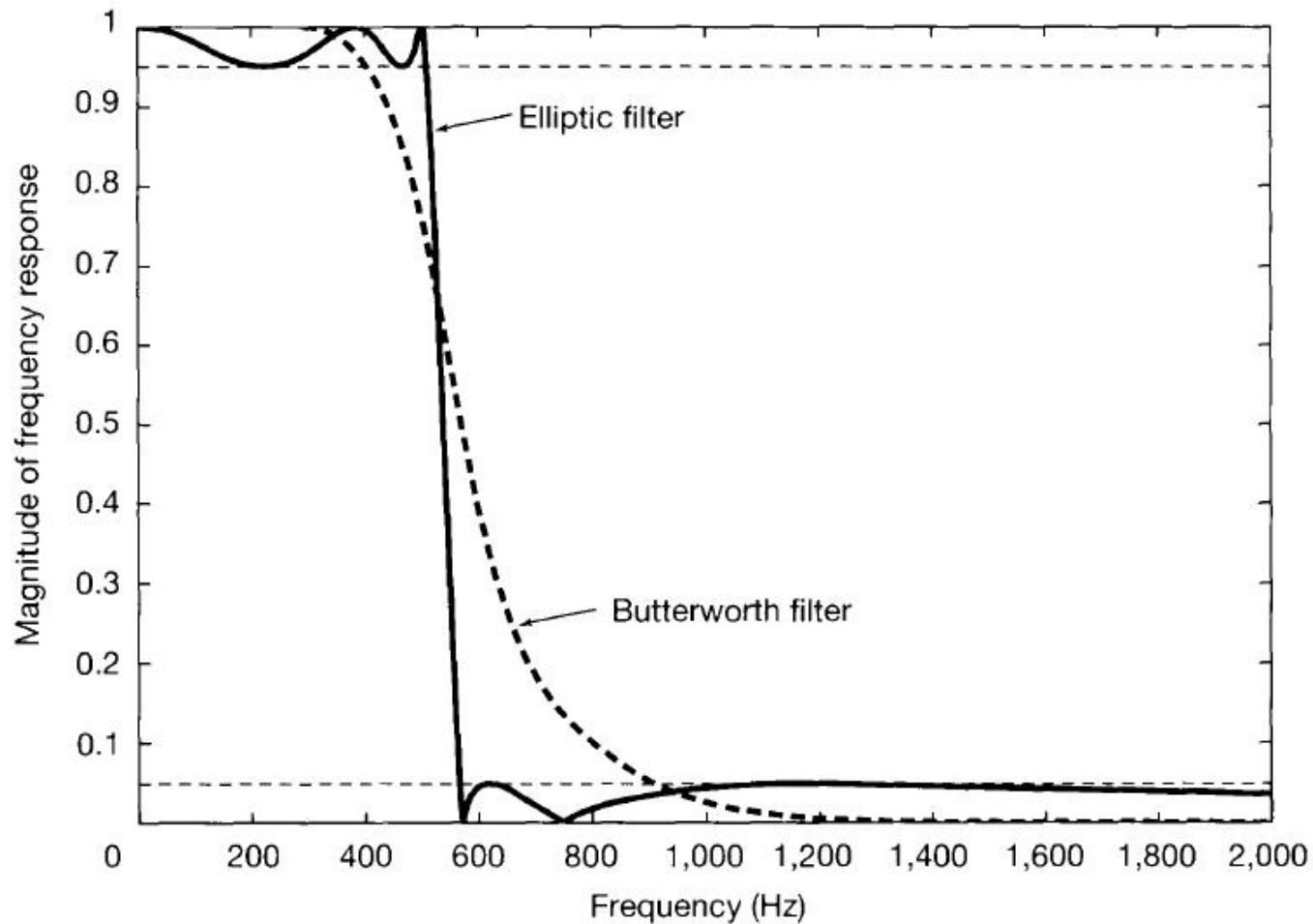


Figure 6.17 Step response of a continuous-time lowpass filter, indicating the rise time t_r , overshoot Δ , ringing frequency ω_r , and settling time t_s —i.e., the time at which the step response settles to within $\pm\delta$ of its final value.

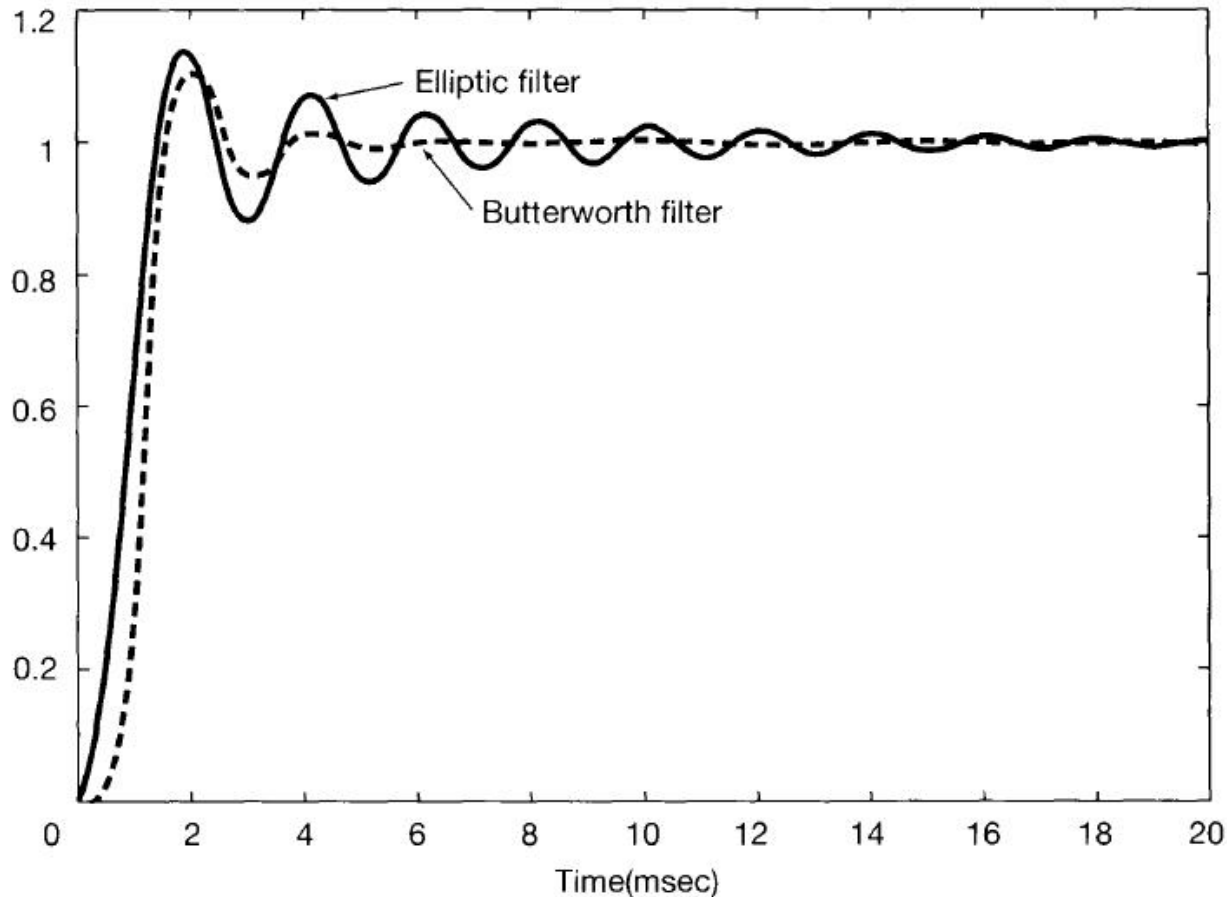
Example:

Elliptic Filter vs. Butterworth Filter



- Transition band: Butterworth filter > Elliptic filter

Cont. – Step response

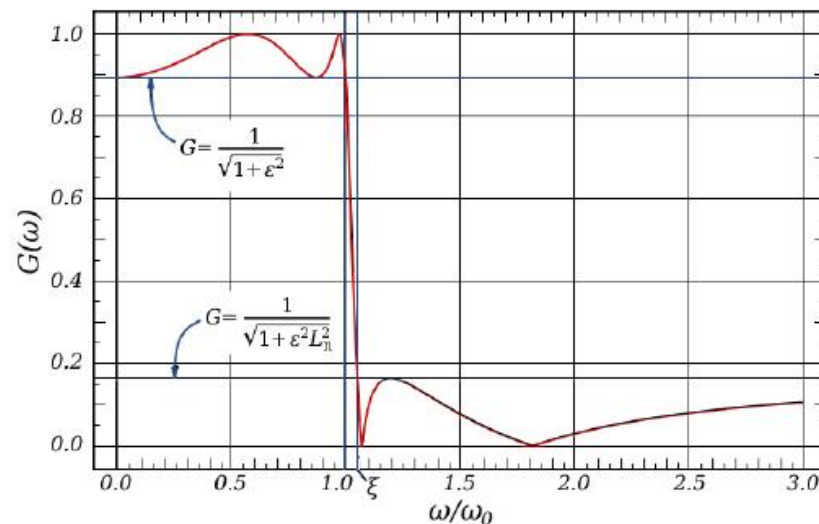


- The ringing in the elliptic filter's step response is more prominent than for the Butterworth step response.

Elliptic Filter

$$G_n(j\omega) = \frac{1}{\sqrt{1 + \epsilon^2 R_n^2(\xi, \omega/\omega_0)}}$$

- where R_n is the n th-order elliptic rational function, ω_0 is the cut-off frequency, ϵ is the ripple factor, ξ is the selectivity factor
- "No other filter of equal order can have a faster transition in gain between the passband and the stopband, for the given values of ripple"



Butterworth Filter

$$G_n(j\omega) = \sqrt{\frac{1}{1 + \omega^{2n}}}$$

- "have as flat a frequency response as possible in the passband"
- Maximally flat magnitude filter

